

Question ID: 98f735f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

The total revenue from sales of a product can be calculated using the formula $T = PQ$, where T is the total revenue, P is the price of the product, and Q is the quantity of the product sold. Which of the following equations gives the quantity of product sold in terms of P and T ?

Answer

A. $Q = \frac{P}{T}$

B. $Q = \frac{T}{P}$

C. $Q = PT$

D. $Q = T - P$

Correct Answer: B

Rationale

Choice B is correct. Solving the given equation for Q gives the quantity of the product sold in terms of P and T. Dividing both sides of the given equation by P yields $\frac{T}{P} = Q$, or $Q = \frac{T}{P}$. Therefore, $Q = \frac{T}{P}$ gives the quantity of product sold in terms of P and T.

Choice A is incorrect and may result from an error when dividing both sides of the given equation by P. Choice C is incorrect and may result from multiplying, rather than dividing, both sides of the given equation by P. Choice D is incorrect and may result from subtracting P from both sides of the equation rather than dividing both sides by P.